

The Sinkhorn Reference Solution and How One Fit Serves Many Grids

Optimal Transport / Schrödinger Bridge Project
Documents `sinkhorn.py` and `study_utils.reference_solution`

Abstract

How the Schrödinger-bridge reference is computed (Hopf–Cole factorisation plus a Sinkhorn iteration against a Neumann heat semigroup), how the resulting trajectory is evaluated, and how a single fine-grid fit is reused as the reference for every LADMM resolution. The last part is the delicate one: the fit lives on one spatial grid and the solver on another, and the reduction between them is not a free choice. Section 5 shows that exact subsampling is *geometrically impossible* for a dyadic sweep of cell-centred grids, that the block average previously used disagreed with the solver’s own marginals at $O(h^2)$, and that this disagreement appeared as a spurious error floor at $t = 0$ and $t = 1$ — amplified to $O(h^2/\Delta t)$, i.e. *growing* under time refinement, whenever it was measured through the FP residual. The fix is to evaluate the fine grid’s own DCT-II series at whatever points are wanted, which is exact and matches the solver’s marginals to machine precision.

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1 Why this reference exists

`analytical_sb_gaussian` solves the free-space bridge on all of $\mathbb{R}$. That is exact and cheap, but only while the density's spread stays small relative to the domain: once ε is large enough that mass reaches $x = 0$ or $x = 1$, the free-space solution and the solver's reflecting-wall problem are different problems. `sinkhorn.py` solves the bridge that the discretisation actually represents — exact discrete marginals on the bounded domain with Neumann (reflecting) walls — and `study_utils` switches to it above `SINKHORN_VAREPS_THRESHOLD` = 5×10^{-3} .

2 Hopf–Cole and the Schrödinger system

Write the bridge density as a product,

$$\rho(t, x) = \varphi(t, x) \psi(t, x), \quad (1)$$

where the two factors solve heat equations running in opposite directions,

$$\partial_t \varphi = \varepsilon \Delta \varphi, \quad -\partial_t \psi = \varepsilon \Delta \psi, \quad (2)$$

subject to the two marginal conditions

$$\varphi(0, \cdot) \psi(0, \cdot) = \rho_0, \quad \varphi(1, \cdot) \psi(1, \cdot) = \rho_1. \quad (3)$$

Both factors are non-negative; neither is a probability density on its own. The pair (φ, ψ) is determined up to the scaling $(\varphi, \psi) \mapsto (c\varphi, \psi/c)$, which leaves ρ unchanged.

2.1 The momentum field

The flux is not an independent unknown — it follows from (1)–(2). Differentiating (1) and using (2),

$$\partial_t \rho = \psi \partial_t \varphi + \varphi \partial_t \psi = \varepsilon (\psi \Delta \varphi - \varphi \Delta \psi) = \varepsilon \nabla \cdot (\psi \nabla \varphi - \varphi \nabla \psi). \quad (4)$$

The Fokker–Planck constraint $\partial_t \rho + \nabla \cdot m = \varepsilon \Delta \rho$ then forces

$$\nabla \cdot m = \varepsilon \Delta (\varphi \psi) - \varepsilon \nabla \cdot (\psi \nabla \varphi - \varphi \nabla \psi) = \varepsilon \nabla \cdot (\psi \nabla \varphi + \varphi \nabla \psi - \psi \nabla \varphi + \varphi \nabla \psi), \quad (5)$$

so that, up to a divergence-free field excluded by the no-flux walls,

$$\boxed{m = 2\varepsilon \varphi \nabla \psi} \quad (6)$$

On the staggered grid this is evaluated with φ averaged and ψ differenced onto the interior x -nodes $x_{j+1/2} = j\Delta x$,

$$m_{j+1/2} = 2\varepsilon \cdot \frac{\varphi_j + \varphi_{j+1}}{2} \cdot \frac{\psi_{j+1} - \psi_j}{\Delta x}, \quad (7)$$

which is second-order accurate and vanishes at the walls whenever ψ is extended evenly, as the Neumann condition requires.

3 The Neumann heat semigroup

Reflecting walls make the Laplacian’s eigenfunctions cosines on the cell-centred grid, so the semigroup is diagonal in the DCT-II basis. With `grid.py`’s eigenvalues

$$\lambda_j = \frac{2 - 2\cos(j\pi/n_x)}{\Delta x^2}, \quad j = 0, \dots, n_x - 1, \quad (8)$$

the action of $H_t = e^{\varepsilon t \Delta}$ is a multiplication in transform space,

$$\widehat{H_t[f]}(j) = e^{-\varepsilon \lambda_j t} \hat{f}(j), \quad (9)$$

implemented as an orthonormal DCT-II, a scaling, and an inverse (`HeatNeumann.apply_time`). Two consequences matter downstream:

- **t is a continuous parameter.** Nothing fixes a time grid. The trajectory can be evaluated at *any* real $t \in [0, 1]$ at the cost of one transform pair, which is what lets one fit serve every n_t .
- **The basis functions are $\cos(j\pi x)$,** sampled at the grid’s cell centres. The same series can therefore be summed at any x , not only at the grid points — §5 uses exactly this.

The projection `np.maximum(out, 0.0)` in `apply_neumann` clips the round-off-level negative values the transform can produce; φ, ψ are non-negative and are subsequently divided by.

4 The Sinkhorn iteration

Fixing (3) is an alternating-projection (IPF) loop. Storing only the two endpoint factors $\varphi(0)$ and $\psi(1)$, one sweep is

$$\varphi(0) \leftarrow \rho_0 / \psi(0), \quad \varphi(1) \leftarrow H_T[\varphi(0)], \quad (10)$$

$$\psi(1) \leftarrow \rho_1 / \varphi(1), \quad \psi(0) \leftarrow H_T[\psi(1)], \quad (11)$$

terminating on the left-marginal residual $\sqrt{\Delta x} \|\varphi(0)\psi(0) - \rho_0\|_2 < \text{tol}$.

One full-time kernel, not n_t steps. Each sweep applies H_T for the *whole* horizon $T = 1$ in a single shot. This is deliberate: $\varphi(0)$ and $\psi(1)$ are narrow (each is a marginal divided by a slowly-varying factor), and one application of the exact semigroup to a narrow function is accurate at any ε , whereas marching in n_t small steps would compound truncation error. It also means the fit never sees n_t or Δt — `sinkhorn_fit` reads only $n_x, \Delta x, \rho_0, \rho_1, \lambda_x$.

Fast convergence at large ε is correct, not degenerate. The iteration typically converges in 2–7 sweeps for $\varepsilon \gtrsim 10^{-2}$, reporting residuals near 10^{-15} . This is the expected behaviour: the contraction rate of an IPF loop improves as the kernel mixes more, and $e^{-\varepsilon \pi^2} = 5.2 \times 10^{-5}$ at $\varepsilon = 1$ means H_T is close to a rank-one averaging operator, for which the fixed point is reached almost immediately. It is worth stating plainly because the symptom invites the opposite reading: a very fast “converged” is easily mistaken for a trivial fixed point. It is not one — the resulting reference’s *interior* FP residual converges cleanly under refinement (at $\varepsilon = 1$, from 2.5×10^{-4} at $n_t = 512$ to 4.6×10^{-6} at $n_t = 8192$).

5 Reusing one fit across grids

The fit is run once per ε on a fine grid (`SINKHORN_NX_FINE` = 4096) so that the reference carries as little of its own discretisation error as possible; the alternative — fitting on each test grid — would contaminate the measured LADMM error with the reference’s $O(\Delta x^2)$ error at the same resolution. Time costs nothing: the trajectory is evaluated directly at each grid’s own edge times $k\Delta t$ and cell-centre times $(k - \frac{1}{2})\Delta t$. **Space is the problem.**

5.1 The consistency requirement

The solver reproduces its marginals *exactly* at $t = 0$ and $t = 1$: the boundary data is baked into the affine map A , not solved for. So whatever the reference says at the endpoints is compared against ρ_0, ρ_1 as `setup_problem` built them,

$$\rho_0 = \frac{\text{pdf}(x_c)}{\sum \text{pdf}(x_c) \Delta x}, \quad x_c = (i - \tfrac{1}{2})\Delta x, \quad (12)$$

a *pointwise* cell-centre sample. Any reference that discretises the same continuous marginal differently will disagree at the endpoints by the difference of the two conventions, and that disagreement is a floor no amount of LADMM convergence can remove.

5.2 Exact subsampling is impossible

The natural wish is to choose n_x^{fine} so that every test grid’s points are a subset of the fine grid’s. It cannot be done. A cell-centred grid on $[0, 1]$ places its points at

$$x_i = \frac{i - \frac{1}{2}}{n_x} = \frac{2i - 1}{2n_x}, \quad (13)$$

i.e. only at *odd* multiples of $1/(2n_x)$.

Lemma 1. *The cell centres of an n_x -grid and a $2n_x$ -grid share no point.*

Proof. A shared point would need $\frac{2i-1}{2n_x} = \frac{2i'-1}{4n_x}$, i.e. $2(2i-1) = 2i'-1$: an even number equal to an odd one. $\square$

Verified directly for $32 \leftrightarrow 64$, $64 \leftrightarrow 128$ and $128 \leftrightarrow 256$: zero shared points in each case. Coarse centres coincide with fine centres exactly when the factor n_x^{fine}/n_x is *odd*, and since $\text{factor}(2n_x) = \text{factor}(n_x)/2$, at most one n_x in a doubling sweep can qualify. At $n_x^{\text{fine}} = 4096$ every factor for $n_x \in \{32, 64, 128, 256\}$ is even, and no other choice helps.

The staggered x -grid is the opposite case and needs no work: m lives at $x_j = j\Delta x$, and *node* grids *are* nested, so `_coarsen.mx` subsamples exactly by picking the aligned fine point. Only ρ is affected.

5.3 The trigonometric interpolant

Let $N = n_x^{\text{fine}}$ and let the fine cell centres be

$$\xi_m = \frac{m - \frac{1}{2}}{N}, \quad m = 1, \dots, N. \quad (14)$$

The functions $\cos(j\pi x)$, $j = 0, \dots, N-1$, are orthogonal with respect to the discrete inner product $\langle f, g \rangle = \sum_m f(\xi_m)g(\xi_m)$, and with the normalisation

$$\beta_0 = \frac{1}{\sqrt{N}}, \quad \beta_j = \sqrt{\frac{2}{N}} \quad (j \geq 1), \quad (15)$$

they are orthonormal. The DCT-II coefficients of a grid function are therefore

$$\hat{f}_j = \beta_j \sum_{m=1}^N f(\xi_m) \cos(j\pi \xi_m), \quad j = 0, \dots, N-1, \quad (16)$$

and the inverse transform recovers the samples,

$$f(\xi_m) = \sum_{j=0}^{N-1} \beta_j \hat{f}_j \cos(j\pi \xi_m). \quad (17)$$

The essential observation is that the right-hand side of (17) is a finite cosine series in x , and a finite cosine series is defined at every point of $[0, 1]$, not only at the ξ_m . Define

$$\boxed{(\mathcal{I}f)(x) := \sum_{j=0}^{N-1} \beta_j \hat{f}_j \cos(j\pi x), \quad x \in [0, 1].} \quad (18)$$

Fact 1. $\mathcal{I}f$ is the unique element of $V_N := \text{span}\{\cos(j\pi x)\}_{j=0}^{N-1}$ with $(\mathcal{I}f)(\xi_m) = f(\xi_m)$ for every m .

Proof. $\dim V_N = N$ and there are N collocation conditions; the matrix of the conditions is $[\beta_j \cos(j\pi \xi_m)]_{jm}$, which is orthogonal by the discrete orthogonality above, hence invertible. $\square$

So $\mathcal{I}$ is interpolation, not smoothing or averaging: it reproduces the fine data exactly and interpolates between the fine points with the same cosines the problem is already written in. The reduction to a coarse grid is then simply evaluation,

$$\rho_i^{(n_x)} = (\mathcal{I}\rho^{\text{fine}})(x_i), \quad x_i = \frac{i - \frac{1}{2}}{n_x}, \quad (19)$$

and (19) places no constraint whatsoever on the relation between n_x and N — in particular it needs no shared points, so the obstruction of §5 simply does not arise.

5.4 Why this is the natural operator here

Two reasons, beyond the absence of a parity condition.

First, V_N is the eigenspace of the very operator that generates the trajectory. The semigroup acts diagonally,

$$H_t[\cos(j\pi \cdot)] = e^{-\varepsilon \lambda_j t} \cos(j\pi \cdot), \quad (20)$$

so each Hopf–Cole factor is, at every time, an element of V_N with known coefficients. Evaluating it at an arbitrary x through (18) is the exact spatial counterpart of evaluating it at an arbitrary t through the factor $e^{-\varepsilon \lambda_j t}$: both are exact evaluations of the same finite representation, and neither introduces a new approximation.

Remark 1. *Strictly, this exactness applies to φ and ψ individually. Their product $\rho = \varphi\psi$ lies in V_{2N} rather than V_N , so applying $\mathcal{I}$ to samples of ρ interpolates rather than reproduces it, with an aliasing error governed by the cosine coefficients of ρ beyond index N . Applying $\mathcal{I}$ to the two factors and multiplying afterwards avoids even that. For the densities of interest the distinction is immaterial: with $N = 4096$ and $\sigma = 0.05$ the relevant coefficient tail is of size $e^{-(N\pi\sigma)^2/2}$, and the measured agreement with the analytic marginal is at the level of 10^{-15} .*

Second, the accuracy is controlled by a coefficient tail rather than by a power of the grid spacing. If c_j denote the true cosine coefficients of f under even extension, then

$$\|f - \mathcal{I}f\|_\infty \leq 2 \sum_{j \geq N} |c_j|, \quad (21)$$

so the error inherits the smoothness of f directly. For a Gaussian of width σ the coefficients decay as $|c_j| \sim e^{-(j\pi\sigma)^2/2}$, and (21) is governed by $e^{-(N\pi\sigma)^2/2}$ — for $N = 4096$, $\sigma = 0.05$ an exponent of order -2×10^5 . This is why the spectral column of Table 1 sits at machine precision rather than at any power of h .

5.5 The two alternatives, and what they cost

Both alternatives are legitimate discretisations; the point is that neither agrees with (12), which is what the comparison requires.

Cell averaging. Replacing the point value by the mean over the coarse cell,

$$(\mathcal{A}f)_i = \frac{1}{\Delta x} \int_{x_i - \Delta x/2}^{x_i + \Delta x/2} f(x) dx, \quad (22)$$

and expanding f about x_i gives

$$(\mathcal{A}f)_i = f(x_i) + \frac{\Delta x^2}{24} f''(x_i) + O(\Delta x^4). \quad (23)$$

The discrepancy with the point value is thus $O(\Delta x^2)$ in the *coarse* spacing, with a constant proportional to f'' . For a bump of width σ , $f'' \sim f/\sigma^2$, so the relative error scales as $\Delta x^2/\sigma^2$ — not small when Δx is comparable to σ , which is precisely the regime these test grids occupy.

Two-point interpolation. When the ratio N/n_x is even, each coarse centre falls exactly midway between two fine centres (§5), and the average of those two neighbours is linear interpolation at that midpoint, with error

$$\frac{1}{8} \Delta x_{\text{fine}}^2 f'' + O(\Delta x_{\text{fine}}^4). \quad (24)$$

The contrast with (23) is the whole story: this error is governed by the *fine* spacing $1/N$, not the coarse one. It is therefore both very small and — a useful check on (24) — essentially independent of n_x , which is exactly what the middle column of Table 1 shows: the same value 6.131×10^{-6} at every test resolution.

The averaging column is $34.2 \Delta x^2$, constant to three digits across the sweep, in agreement with (23).

Table 1: Discrepancy between the reduced reference marginal and the solver’s own ρ_1 of (12), for $\sigma = 0.05$ and $N = 4096$. The averaging column is $O(\Delta x^2)$ in the coarse spacing; the interpolation column is $O(N^{-2})$ and hence flat; the spectral column is at machine precision.

n_x	cell average $\mathcal{A}$	two-point interpolation	spectral $\mathcal{I}$
32	3.307e-02	6.131e-06	1.546e-15
64	8.343e-03	6.131e-06	1.432e-15
128	2.089e-03	6.131e-06	2.148e-15
256	5.210e-04	6.131e-06	4.776e-15

5.6 Fitting on the test grid instead

Everything above concerns reducing a fine fit. The alternative is to drop the reduction entirely and fit on the solver’s own spatial grid. The two choices are not competitors; they measure different quantities.

Fitting at n_x^{fine} makes the reference approximately the *continuum* bridge, so the measured discrepancy is the solver’s total discretisation error — what a convergence study needs. The cost is that the reference’s own evolution uses the fine Laplacian while the solver’s uses the coarse one, and the two differ at $O(\Delta x^2)$; that difference is what (25) amplifies at the endpoints.

Fitting at n_x removes that difference by construction. Three things follow. The reduction step disappears, so none of §5 applies. The marginals coincide exactly, because the fit is driven by the solver’s own ρ_0, ρ_1 of (12). And — the useful part — the Fokker–Planck constraint and the Sinkhorn reference then share a spatial discretisation, both being diagonalised by the same λ_j of (8). Since the semigroup is exact in time, the reference’s FP residual must then vanish as $\Delta t \rightarrow 0$ at fixed Δx , leaving only the solver’s *temporal* error. It does, at second order:

n_t ($n_x = 128, \varepsilon = 0.1$)	fine fit, interior	same-grid fit, interior
128	5.360e-04	1.708e-04
512	3.685e-04	1.318e-05
2048	3.584e-04	8.679e-07
8192	3.594e-04	5.498e-08

The fine-fit column plateaus at the fixed spatial error of the $n_x = 128$ grid; the same-grid column converges to zero. The same contrast holds at the $t = 0$ row, where the fine fit’s residual *grows* (1.6×10^{-1} to 4.8×10^{-1} over that sweep, the $O(\Delta x^2/\Delta t)$ term of (25)) while the same-grid fit’s falls to 6.2×10^{-5} .

The limitation is the mirror image of the benefit: a reference that is spatially consistent with the solver by construction is *blind* to spatial error, and cannot support any claim about convergence in Δx . Use the fine fit for convergence studies and the same-grid fit for isolating the time discretisation.

6 The failure this caused

`coarsen_rho` was the original behaviour, and its $O(\Delta x^2)$ mismatch produced a spurious error floor at *both* endpoints — present only above `SINKHORN_VAREPS_THRESHOLD`, since below it the analytic reference uses the same pointwise convention as (12) and no mismatch arises.

The effect is much worse when the reference is inspected through the FP residual rather than through $\|\rho - \rho_{\text{ref}}\|$, because the terminal constraint row divides by Δt :

$$\text{last-row residual} \sim \underbrace{\frac{34.2 \Delta x^2}{\Delta t}}_{\text{convention mismatch}} + \underbrace{B(\varepsilon) \Delta t}_{\text{scheme}}. \quad (25)$$

The first term *grows* as $\Delta t \rightarrow 0$ at fixed Δx . Measured at $\varepsilon = 10^{-2}$, $n_x = 256$: the last-row residual ran $0.063 \rightarrow 0.124 \rightarrow 0.257 \rightarrow 0.523$ for $n_t = 128, 256, 512, 1024$ — a clean factor 2 per halving of Δt , and (25) predicts $34.2 \cdot 256^{-2} \cdot 1024 = 0.53$ against the observed 0.523. Switching to either non-averaging reduction removes it:

ε	n_t ($n_x = 128$)	"average"	"pointwise"	"spectral"
0.01	128	2.326e-01	4.323e-02	4.332e-02
0.01	512	1.029e+00	2.311e-02	2.351e-02
0.01	2048	4.237e+00	2.351e-02	2.259e-02

"average" diverges; the other two converge. Note also that spectral and pointwise are numerically indistinguishable in this test — the 6×10^{-6} interpolation error is far below everything else. The fix that mattered was *away from block averaging*; spectral is preferred because it is the principled construction (no operator, any n_x , exact), not because it moves these numbers.

What it does not explain. The second term of (25) survives every reduction: at $\varepsilon = 1$, $n_t = 2048$ the last-row residual is 4.18×10^1 under all three. That term is a property of the ETD terminal row, not of the reference, and is analysed in `etd_theta_projection.tex` §9.

7 Practical guidance

- The three reductions of §5 are selected by `reference_solution`'s `rho_reduce` argument, with values "spectral" ($\mathcal{I}$, the default, eq. (19)), "pointwise" (two-point interpolation, eq. (24)) and "average" ($\mathcal{A}$, eq. (23)). Leave it at the default; use "average" only to reproduce pre-fix numbers.
- `SINKHORN_NX_FINE` must be an integer multiple of every n_x in the sweep. It no longer needs any parity property, and is irrelevant under the same-grid option of §5.6.
- The two fitting grids are selected by `study_utils.sinkhorn_reference_fit`'s `grid` argument ("fine" or "same"), surfaced as `--sinkhorn-grid` in `sb_gaussian.py` and `sb_gaussian_nonuniform.py`. The same-grid option is fitted per resolution by necessity, which is cheap: the iteration needs only a handful of sweeps.
- Fit once per ε , not once per resolution — `sinkhorn_fit` does not depend on n_t , and `reference_solution` takes the fit as an argument precisely so it can be shared. Note that `run_study` refits internally per call, so comparing two schemes through it repeats the fine-grid solve; `compare_etd1_vs_theta.py` fits once and passes it down instead.
- The reference still carries its own $O(\Delta x_{\text{fine}}^2)$ error. At $n_x^{\text{fine}} = 4096$ against test grids up to 256 that is ~ 256 times smaller than the quantity being measured, but it is not zero, and it is the floor on any convergence study run against it.

- Below `SINKHORN_VAREPS_THRESHOLD` the analytic reference is used and none of §5 applies; it is evaluated directly on the test grid.